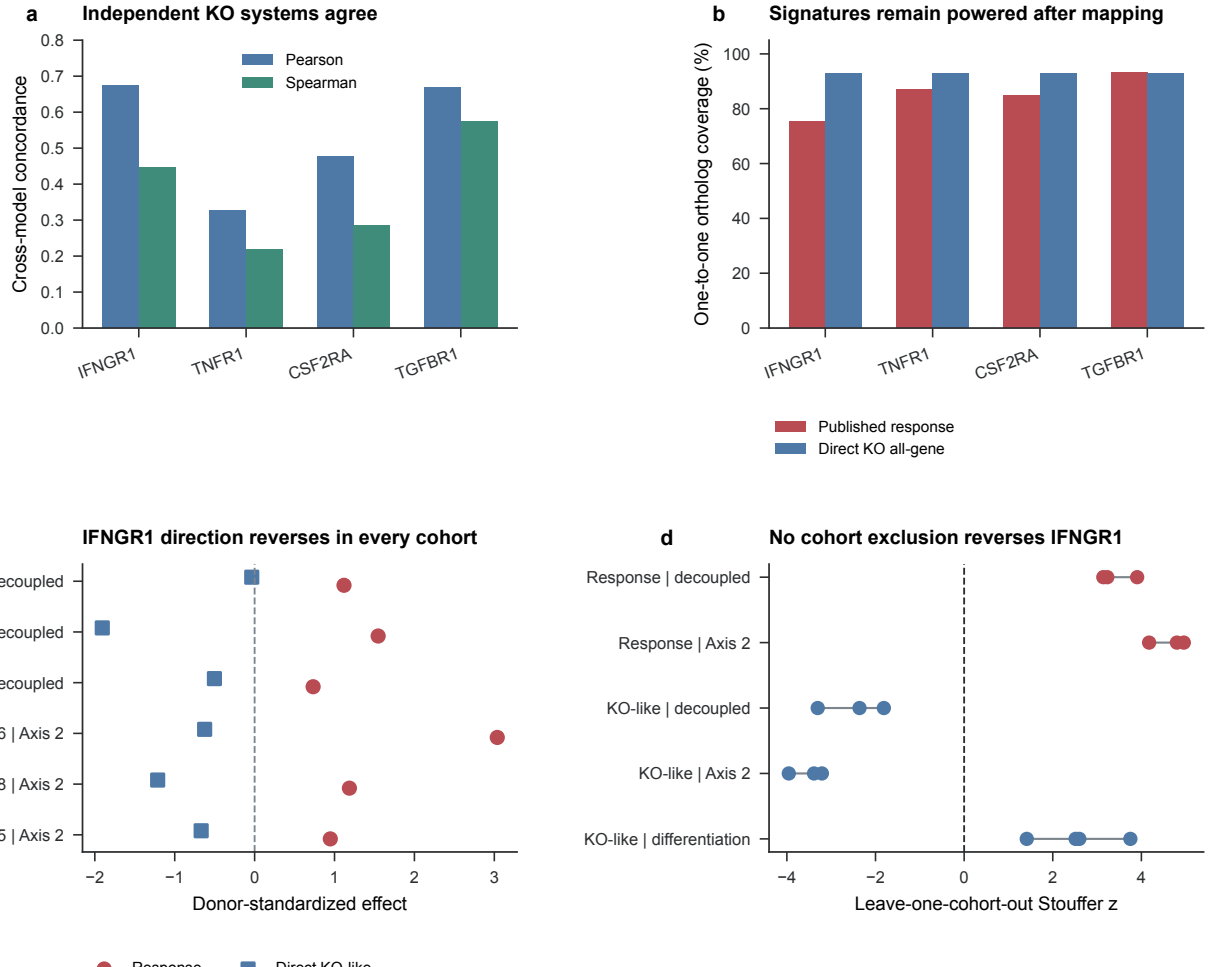
a
Independent KO systems agree
Cross-model concordance
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Pearson
Spearman
IFNGR1
TNFR1
CSF2RA
TGFBR1
b
Signatures remain powered after mapping
One-to-one ortholog coverage (%)
0
20
40
60
80
100
IFNGR1
TNFR1
CSF2RA
TGFBR1
Published response
Direct KO all-gene
c
IFNGR1 direction reverses in every cohort
138266 | decoupled
286068 | decoupled
328265 | decoupled
138266 | Axis 2
286068 | Axis 2
328265 | Axis 2
−2
−1
0
1
2
3
Donor-standardized effect
Response
Direct KO-like
d
No cohort exclusion reverses IFNGR1
Response | decoupled
Response | Axis 2
KO-like | decoupled
KO-like | Axis 2
KO-like | differentiation
−4
−2
0
2
4
Leave-one-cohort-out Stouffer z